EDUX / ACADEMIC INTELLIGENCE

FULL EXAM ANALYSIS REPORT

QA Stress Fixture — variable subjects, scores and question counts

Synthetic validation paper · Mixed format · All Subjects · Generated 02 Nov 2026

STUDENT	ROLL NUMBER	EXAMINATION DATE	TOTAL MARKS
QA Fixture	QA-0001	2026-11-02	360

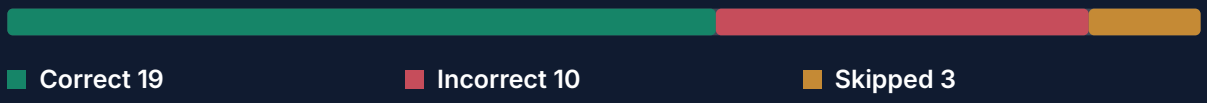
TOTAL MARKS OBTAINED

217

/ 360 marks

19 correct | 10 incorrect | 3 skipped





Correct 19 Incorrect 10 Skipped 3

ACCURACY	PERCENTILE	RANK	AVERAGE PACE
66.7%	88.2	#91	186m 12s

PUBLISHED PERFORMANCE BRIEF

This is a synthetic edge-case fixture used only to validate pagination, mathematical typesetting, long labels, missing optional values and dynamic subject combinations. It is not a student report.

01 PERFORMANCE INTELLIGENCE

Performance intelligence

Response composition

32 questions in the paper



- Correct 19
- Incorrect 10
- Skipped 3

Difficulty analysis

Questions and published accuracy

Easy		10 Q · 82%
Medium		12 Q · 66%
Hard		10 Q · 41%

Accuracy values are reproduced from the payload.

Mistake distribution

Count by published classification

Concept Error		5
Calculation Error		4
Formula Error		3
Time Pressure		2

Source classifications; rows are measured independently.

Subject-wise performance

Subject-wise performance

Accuracy by subject



Physics 121 / 180 Chemistry 96 / 180

Important performance signals

GRADE **B**	RANK **91**	PERCENTILE **88.2**
BATCH RANK **4**	COHORT SIZE **60**	OVERALL SPEED **2.8**
CONFIDENCE INDEX **74**	READINESS SCORE **70**	HEALTH SCORE **79**
NEGATIVE MARKS **9**	GUESS ATTEMPTS **2**	ATTEMPT RATIO **90.7%**
TOTAL QUESTIONS **32**	SUCCESS RATE **66.7%**	BADGE **Synthetic QA**

02 MASTERY MAP

Subject and chapter mastery

Subject marks, accuracy and classification

SUBJECT	MARKS	ACCURACY	RANK	LEVEL	STRONG AREAS	WEAK AREAS
Physics	121 / 180	67%	2	Medium	Kinematics	Rotational Motion
Chemistry	96 / 180	53%	4	Hard	Mole Concept	Chemical Kinetics

Strong, weak and critical chapters by subject

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Physics	—	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Rotational Motion 5	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Rotational Motion 5
Chemistry	Chemical Kinetics 4 · Chemical Kinetics 8	Chemical Kinetics 2 · Chemical Kinetics 6	—

Chapter mastery · grouped by subject

CHAPTER	ACCURACY	MARKS	TIME	MASTERY
Physics				
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	42%	4	3m	Critical
Chemistry				
Chemical Kinetics 2	48%	5	4m	Weak
Physics				
Rotational Motion 3	54%	6	5m	Average
Chemistry				
Chemical Kinetics 4	60%	7	6m	Strong
Physics				
Rotational Motion 5	66%	8	7m	Critical
Chemistry				
Chemical Kinetics 6	72%	9	8m	Weak

Chapter mastery · grouped by subject · continued

CHAPTER	ACCURACY	MARKS	TIME	MASTERY
Physics				
Rotational Motion 7	78%	10	9m	Average
Chemistry				
Chemical Kinetics 8	84%	11	10m	Strong

03 PACE INTELLIGENCE

Time intelligence

Chapter time consumption · measured rows

CHAPTER	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Physics			
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	6%	3m	6%
Chemistry			
Chemical Kinetics 2	8%	4m	8%
Physics			
Rotational Motion 3	10%	5m	10%
Chemistry			
Chemical Kinetics 4	12%	6m	12%
Physics			
Rotational Motion 5	13%	7m	13%
Chemistry			
Chemical Kinetics 6	15%	8m	15%
Physics			
Rotational Motion 7	17%	9m	17%
Chemistry			
Chemical Kinetics 8	19%	10m	19%

Time-management metrics

AVERAGE PER QUESTION
3

NAVIGATION COUNT
11

MANAGEMENT SCORE
68

FASTEST QUESTION
Q1 — Kinematics

FASTEST TIME
0.7

SLOWEST QUESTION
Q31 — Definite Integration

SLOWEST TIME
7.9

Sectional allocation and efficiency

SECTION	ALLOCATED	USED	EFFICIENCY
Physics	60	48	125
Chemistry	60	42	143

04 COMPARISON & TRAJECTORY

Comparison and projection

Comparative performance

Comparison register

BENCHMARK	SCORE	PERCENTILE	GRADE	REFERENCE
Previous test	198	84.1	B	—
Batch average	176	80.2	—	—

VS PREVIOUS TEST
+19

Projection and readiness

Score trajectory



Expected improvement +3.1% · Risk Medium

JEE PERCENTILE	NEET SCORE	EXPECTED AIR
91.1	—	21400
TARGET PROBABILITY	**RISK LEVEL**	**EXPECTED IMPROVEMENT**
77	Medium	+3.1%

Trajectory values

EXAMINATION / PROJECTION	SCORE
QA 1	190
QA 2	217

READINESS SCORE	CONFIDENCE INDEX	HEALTH SCORE
70	74	79
OVERALL ACCURACY	**SUCCESS RATE**	**BADGE**
66.7%	66.7%	Synthetic QA

05 QUESTION INTELLIGENCE

Question-by-question analysis

32 questions reviewed · published explanations reproduced in full

01 Chemistry

11 questions

QUESTION 01 CHEMISTRY

CORRECT

Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Easy · MCQ · 4 marks

Evaluate the expression for case 1: $x^{1} + \sqrt{1} + \sum_{k=1}^{1} k$.

STUDENT RESPONSE
Not published

CORRECT ANSWER
Not published

PERFORMANCE INSIGHT
Answered correctly — 4 marks · Used u = x^2 instead of u = x^2+1

EXPLANATION

The integral reduces directly:

$$\int_{0}^{1} \frac{x^{2}}{\sqrt{1-x^{2}}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

02 Physics

21 questions

QUESTION 02 PHYSICS

INCORRECT

Rotational Motion · Projectile motion · Medium · Numerical · −1 mark · 1m 54s

Evaluate the expression for case 2: $x^2 + \sqrt{2} + \sum_{k=1}^{2} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Physics · Q3 · question continued

QUESTION 03 PHYSICS

SKIPPED

Rotational Motion · Projectile motion · Hard · Assertion Reason · 4 marks · 2m 36s

Evaluate the expression for case 3: $x^3 + \sqrt{3} + \sum_{k=1}^{3} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

03 Chemistry

11 questions

QUESTION 04 CHEMISTRY

CORRECT

Chemical Kinetics · Projectile motion · Easy · MCQ · 4 marks · 3m 18s

Evaluate the expression for case 4: $x^4 + \sqrt{4} + \sum_{k=1}^{4} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

04 Physics

21 questions

QUESTION 05 PHYSICS

INCORRECT

Rotational Motion · Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Medium · Numerical · −1 mark · 4m

Evaluate the expression for case 5: $x^5 + \sqrt{5} + \sum_{k=1}^{5} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Physics · Q6 · question continued

QUESTION 06 PHYSICS

SKIPPED

Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Projectile motion · Hard · Assertion Reason · 4 marks · 4m 42s

Evaluate the expression for case 6: $x^{6} + \sqrt{6} + \sum_{k=1}^{6} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Using

$$\sum_{i=1}^{n} i^{2} = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^{2} + \beta^{2} \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

05 Chemistry

11 questions

QUESTION 07 CHEMISTRY

CORRECT

Chemical Kinetics · Projectile motion · Easy · MCQ · 4 marks · 5m 24s

Evaluate the expression for case 7: $x^{7} + \sqrt{7} + \sum_{k=1}^{7} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

06 Physics

21 questions

QUESTION 08 PHYSICS

INCORRECT

Rotational Motion · Projectile motion · Medium · Numerical · −1 mark

Evaluate the expression for case 8: $x^{8} + \sqrt{8} + \sum_{k=1}^{8} k$.

STUDENT RESPONSE
Not published

CORRECT ANSWER
Not published

PERFORMANCE INSIGHT
Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

Physics · Q9 · question continued

QUESTION 09 PHYSICS

SKIPPED

Rotational Motion · Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Hard · Assertion Reason · 4 marks · 6m 48s

Evaluate the expression for case 9: $x^{9} + \sqrt{9} + \sum_{k=1}^{9} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The integral reduces directly:

$$\int_{0}^{1} \frac{x^{2}}{\sqrt{1-x^{2}}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

07 Chemistry

11 questions

QUESTION 10 CHEMISTRY

CORRECT

Chemical Kinetics · Projectile motion · Easy · MCQ · 4 marks · 1m 12s

Evaluate the expression for case 10: $x^{10} + \sqrt{10} + \sum_{k=1}^{10} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered correctly — 4 marks · Used u = x^2 instead of u = x^2+1

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

08 Physics

21 questions

QUESTION 11 PHYSICS

INCORRECT

Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Projectile motion · Medium · Numerical · −1 mark · 1m 54s

Evaluate the expression for case 11: $x^{11} + \sqrt{11} + \sum_{k=1}^{11} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Physics · Q12 · question continued

QUESTION 12 PHYSICS

SKIPPED

Rotational Motion · Projectile motion · Hard · Assertion Reason · 4 marks · 2m 36s

Evaluate the expression for case 12: $x^{12} + \sqrt{12} + \sum_{k=1}^{12} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

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EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

09 Chemistry

11 questions

QUESTION 13 CHEMISTRY

CORRECT

Chemical Kinetics · Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Easy · MCQ · 4 marks · 3m 18s

Evaluate the expression for case 13: $x^{13} + \sqrt{13} + \sum_{k=1}^{13} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

10 Physics

21 questions

QUESTION 14 PHYSICS

INCORRECT

Rotational Motion · Projectile motion · Medium · Numerical · −1 mark · 4m

Evaluate the expression for case 14: $x^{14} + \sqrt{14} + \sum_{k=1}^{14} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

QUESTION 15 PHYSICS

SKIPPED

Rotational Motion · Projectile motion · Hard · Assertion Reason · 4 marks

Evaluate the expression for case 15: $x^{15} + \sqrt{15} + \sum_{k=1}^{15} k$.

STUDENT RESPONSE
Not attempted

CORRECT ANSWER
Not attempted

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

11 Chemistry

11 questions

QUESTION 16 CHEMISTRY

CORRECT

Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Projectile motion · Easy · MCQ · 4 marks · 5m 24s

Evaluate the expression for case 16: $x^{16} + \sqrt{16} + \sum_{k=1}^{16} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

12 Physics

21 questions

QUESTION 17 PHYSICS

INCORRECT

Rotational Motion · Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Medium · Numerical · −1 mark · 6m 06s

Evaluate the expression for case 17: $x^{17} + \sqrt{17} + \sum_{k=1}^{17} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Physics · Q18 · question continued

QUESTION 18 PHYSICS

SKIPPED

Rotational Motion · Projectile motion · Hard · Assertion Reason · 4 marks · 6m 48s

Evaluate the expression for case 18: $x^{18} + \sqrt{18} + \sum_{k=1}^{18} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

13 Chemistry

11 questions

QUESTION 19 CHEMISTRY

CORRECT

Chemical Kinetics · Projectile motion · Easy · MCQ · 4 marks · 1m 12s

Evaluate the expression for case 19: $x^{19} + \sqrt{19} + \sum_{k=1}^{19} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered correctly — 4 marks · Used u = x^2 instead of u = x^2+1

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

14 Physics

21 questions

QUESTION 20 PHYSICS

INCORRECT

Rotational Motion · Projectile motion · Medium · Numerical · −1 mark · 1m 54s

Evaluate the expression for case 20: $x^{20} + \sqrt{20} + \sum_{k=1}^{20} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

Physics · Q21 · question continued

QUESTION 21 PHYSICS

SKIPPED

Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Hard · Assertion Reason · 4 marks · 2m 36s

Evaluate the expression for case 21: $x^{21} + \sqrt{21} + \sum_{k=1}^{21} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

15 Chemistry

11 questions

QUESTION 22 CHEMISTRY

CORRECT

Chemical Kinetics · Projectile motion · Easy · MCQ · 4 marks

Evaluate the expression for case 22: $x^{22} + \sqrt{22} + \sum_{k=1}^{22} k$.

STUDENT RESPONSE

Not published

CORRECT ANSWER

Not published

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

16 Physics

21 questions

QUESTION 23 PHYSICS

INCORRECT

Rotational Motion · Projectile motion · Medium · Numerical · −1 mark · 4m

Evaluate the expression for case 23: $x^{23} + \sqrt{23} + \sum_{k=1}^{23} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Physics · Q24 · question continued

QUESTION 24 PHYSICS

SKIPPED

Rotational Motion · Projectile motion · Hard · Assertion Reason · 4 marks · 4m 42s

Evaluate the expression for case 24: $x^{24} + \sqrt{24} + \sum_{k=1}^{24} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

17 Chemistry

11 questions

QUESTION 25 CHEMISTRY

CORRECT

Chemical Kinetics · Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Easy · MCQ · 4 marks · 5m 24s

Evaluate the expression for case 25: $x^{25} + \sqrt{25} + \sum_{k=1}^{25} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

A

CORRECT ANSWER

B

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

18 Physics

21 questions

QUESTION 26 PHYSICS

INCORRECT

Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Projectile motion · Medium · Numerical · −1 mark · 6m 06s

Evaluate the expression for case 26: $x^{26} + \sqrt{26} + \sum_{k=1}^{26} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

B

CORRECT ANSWER

C

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Physics · Q27 · question continued

QUESTION 27 PHYSICS

SKIPPED

Rotational Motion · Projectile motion · Hard · Assertion Reason · 4 marks · 6m 48s

Evaluate the expression for case 27: $x^{27} + \sqrt{27} + \sum_{k=1}^{27} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Not attempted — no marks awarded

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

19 Chemistry

11 questions

QUESTION 28 CHEMISTRY

CORRECT

Chemical Kinetics · Projectile motion · Easy · MCQ · 4 marks · 1m 12s

Evaluate the expression for case 28: $x^{28} + \sqrt{28} + \sum_{k=1}^{28} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered correctly — 4 marks · Used u = x^2 instead of u = x^2+1

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

20 Physics

21 questions

QUESTION 29 PHYSICS

INCORRECT

Rotational Motion · Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions · Medium · Numerical · −1 mark

Evaluate the expression for case 29: $x^{29} + \sqrt{29} + \sum_{k=1}^{29} k$.

STUDENT RESPONSE
Not published

CORRECT ANSWER
Not published

PERFORMANCE INSIGHT
Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

QUESTION 30 PHYSICS

SKIPPED

Rotational Motion · Projectile motion · Hard · Assertion Reason · 4 marks · 2m 36s

Evaluate the expression for case 30: $x^{30} + \sqrt{30} + \sum_{k=1}^{30} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE
B

CORRECT ANSWER
C

PERFORMANCE INSIGHT
Not attempted — no marks awarded

EXPLANATION

Using

Physics · Q30 · explanation continued

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

21 Chemistry

11 questions

QUESTION 31 CHEMISTRY

CORRECT

Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Projectile motion · Easy · MCQ · 4 marks · 3m 18s

Evaluate the expression for case 31: $x^{31} + \sqrt{31} + \sum_{k=1}^{31} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

C

CORRECT ANSWER

D

PERFORMANCE INSIGHT

Answered correctly — 4 marks

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION HINT

Check the sign convention and the limits of integration carefully.

22 Physics

21 questions

QUESTION 32 PHYSICS

INCORRECT

Rotational Motion · Projectile motion · Medium · Numerical · −1 mark · 4m

Evaluate the expression for case 32: $x^{32} + \sqrt{32} + \sum_{k=1}^{32} k$.

OPTIONS

A. 2 s

B. $\sqrt{3}$ s

C. 4 s

D. A very long alternative answer that must wrap without clipping the card

STUDENT RESPONSE

D

CORRECT ANSWER

A

PERFORMANCE INSIGHT

Answered incorrectly — 1 mark deducted · Concept Error

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

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```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

06 LEARNING INTELLIGENCE

Learning intelligence

Published revision priorities

Priority revision register

PRIORITY TOPIC	CLASSIFICATION	TIMEFRAME
Definite Integration — substitution	Critical	48 hours
Chemical Kinetics — rate laws	High	5 days

Topic-level accuracy

TOPIC	SUBJECT	ACCURACY	LEVEL
Definite Integration	Physics	44%	Critical
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Physics	51%	Weak
Bayes theorem	Physics	62%	Average
Coordination compounds	Chemistry	48%	Weak
Chemical Kinetics	Chemistry	82%	Strong
Mole Concept	Chemistry	91%	Strong

Mistake evidence

QUESTION	SOURCE DETAIL
Q4	Misapplied substitution — used u = x^2 instead of u = x^2+1
Q11	Decimal error in 1/r^2 computation

Weak chapters

NO.	WEAK CHAPTER
1	Calculus (44%)
2	Chemical Kinetics (48%)

Subject-wise revision sections

Priority topics grouped by subject

PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Physics			
Definite Integration	—	44%	Critical
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	51%	Weak
Bayes theorem	—	62%	Review
Chemistry			
Coordination compounds	—	48%	Weak

Recommended next actions

Published action register

ACTION	RESOURCE	DETAIL
PYQ	**JEE Main 2024 — Integration (5 Qs)**	PYQ Bank
Practice	**20-question drill — substitution**	20 · Medium
Mock test	**Sectional mock — Chemistry**	Sectional
Lecture	**Kinetics masterclass**	Video

Weak topics · published list

NO.	WEAK TOPIC
1	**Definite Integration**
2	**Coordination compounds**
3	**Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions**

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End of report · 02 Nov 2026